Ising Model

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Lenz imagined a magnet as a lattice of little arrows, each pointing either up or down, representing atoms. (Atoms are intrinsically magnetic, with north and south poles, and thus can be thought of as having orientations.) Arrows influence their neighbors, attempting to magnetically flip them to match their own orientation.

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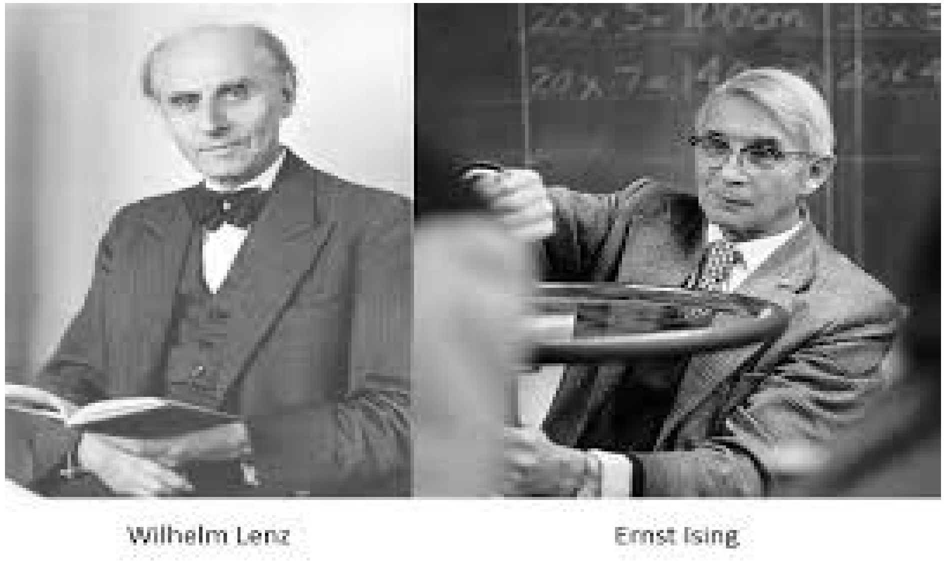
The feature that launched thousands of physics papers was the struggle within the lattice between heat and magnetism. Heat, the random jiggling of particles, fought for disorder; magnetism resisted this chaos.

Lenz supposed that at low temperatures, magnetic order should win. With enough heat, however, random jostling would disrupt the atomic cooperation, explaining Curie’s observation that hot magnets lose their mojo.

Lenz tasked his graduate student, Ernst Ising, with working out the details. Though real magnets are three-dimensional, Ising simplified the situation to a linear chain of arrows, each capable of sensing its two nearest neighbors. Through a calculation now standard in undergraduate textbooks, he showed that the chain — now known as the 1D Ising model — fails to stay magnetized.
Random flips overwhelm magnetic cohesion at all temperatures.

Ising published his findings in 1925. He and Lenz assumed that the results held for 2D sheets and 3Dblocks of arrows as well, and that the model therefore failed to capture the behavior of real magnets. It seemed destined for the graveyard of incorrect physical theories.

No phase transition, no criticality



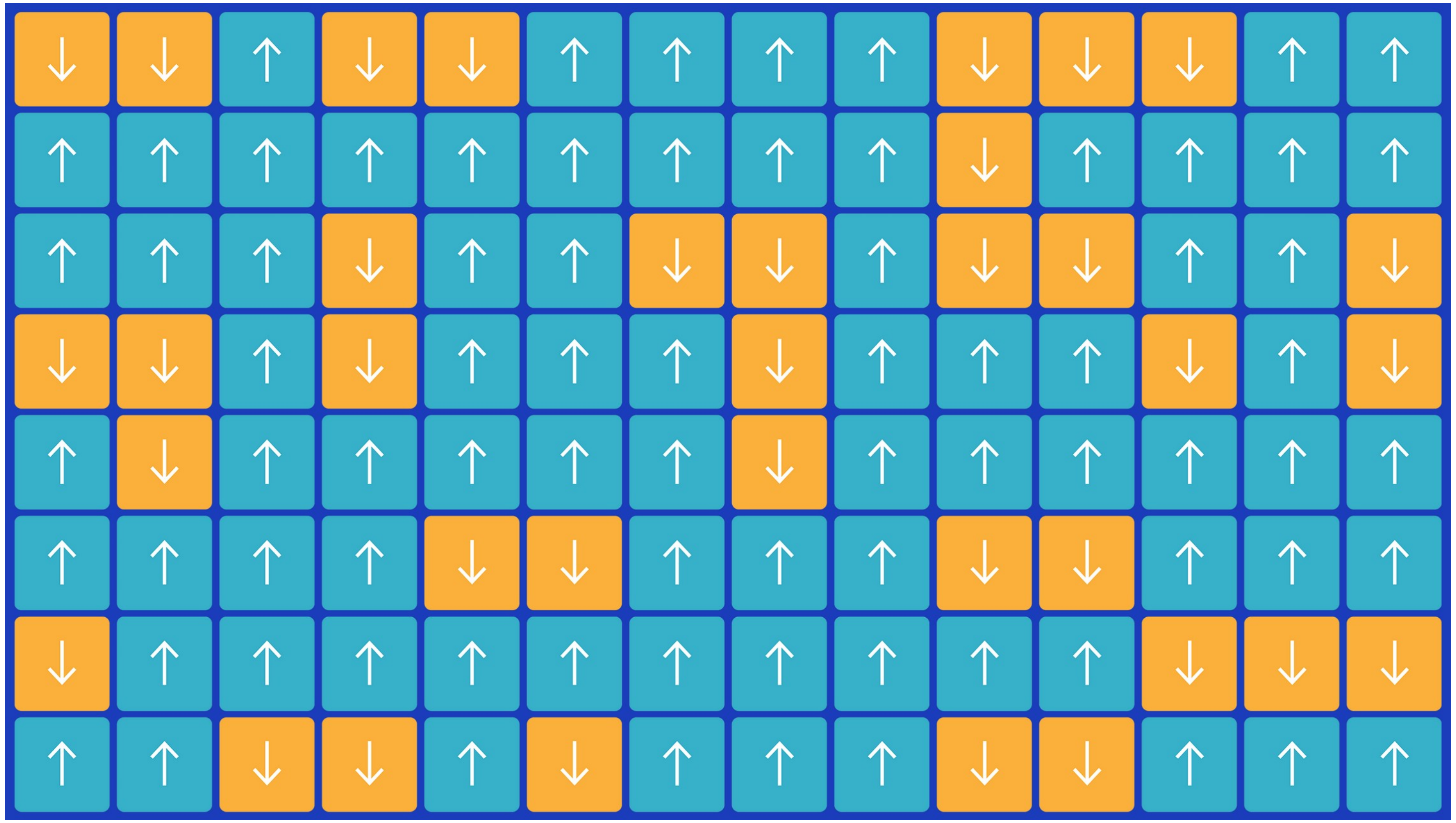
8 February 1888 30 April 1957

May 10, 1900 – May 11, 1998

Onsager’s proof showed that in 2D, just as Lenz had suspected, arrows align and magnetism carries the day at low temperatures, while disorder wins out after the system surpasses a “critical temperature.” A simple grid of interlinked arrows explains phase transitions. There was no need to incorporate the messiness of real particles, as many physicists had supposed.

The American physicist Ken Wilson worked out the mathematics of universality in 1971, winning a Nobel Prize for the effort. Wilson showed that, whereas at high temperatures arrows point which ever way they like, as the system cools and approaches its phase transition, magnetic attraction between neighbors creates larger and larger “islands” of order where all arrows point together. Critical exponents describe the details of this process, such as how the biggest islands grow.

Ising Model: Motivation



https://www.bdhammel.com/ising-model/
https://www.quantamagazine.org/the-cartoon-picture-of-magnets-that-has-transformed-science-20200624/

https://stanford.edu/~jeffjar/statmech/lec3.html

Ising Model: Motivation

Why should we spend so much time talking about the Ising model?

•*It's surprisingly* **useful for helping us think** *about all sorts of behaviors relating to phase transitions. For instance:*

- *It exhibits symmetry breaking in low-temperature phase .*
- *'Critical point' at a well-defined temperature*

•It's one of few **exactly solvable models** where we can actually compute thermodynamic quantities and interpret them. In general, calculating thermodynamic quantities is hard because

- You have to sum up many terms. Remember that you can think about an equilibrium system as an *ensemble* of many states s , each weighted with their own probability P_s.

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Ising Model: Motivation, Background

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- In this framework, the thermodynamic quantities that you observe correspond to averages over the ensemble. In particular, if you want to find the ensemble average of some observable $A(s)$, you need to find the sum $\langle A \rangle = \sum_s A(s)P(s)$ where the sum runs over *all the possible states*.

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Statistical Physics Seminar by Prof. Wolschin
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- ➢ You have to sum up many terms. Remember that you can think about an equilibrium system as an *ensemble* of many states s , each weighted with their own probability P_s .
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- ➢ The problem with this, as you remember, is that the number of states of a thermodynamic system scales *exponentially* with the number of particles! Even for a moderate-sized system, there are just too many states for a computer to explicitly compute the average – let alone a thermodynamic system where N is on the order of 10^{23} .

•So we need to "be clever" to compute the partition function, and we ought to be thankful for exactly solvable systems!

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Ising Model

- The Ising model is a theoretical model in statistical physics that was originally developed to describe ferromagnetism.
- A system of magnetic particles can be modeled as a linear chain in one dimension or a lattice in two dimension, with one molecule or atom at each lattice site i.

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•An alloy; say, brass. Each of the sites is an atom in the lattice; -1 represents an copper atom at that site; +1 represents zinc.
•A "lattice gas". Each of the sites is the possible location of a particle; -1 means that site is empty and +1 means that site is occupied by a particle.

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Ising Model

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- A system of magnetic particles can be modeled as a linear chain in one dimension or a lattice in two dimension, with one molecule or atom at each lattice site i.
- To each molecule or atom a magnetic moment is assigned that is represented in the model by a discrete variable σ_i. Each 'spin' σ_i can only have a value of $\sigma_i = \pm 1$. The two possible values indicate whether two spins σ_i and σ_j are align and thus parallel ($\sigma_i\ \sigma_j = +1$) or anti-parallel ($\sigma_i\ \sigma_j = -1$).

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Ising Model

- *A system of two spins is considered to be in a lower energetic state if the two magnetic moments are aligned. If the magnetic moments points in opposite directions they are consider to be in a higher energetic state. Due to this interaction the system tends to align all magnetic moments in one direction in order to minimize energy.*

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- *If nearly all magnetic moments point in the same direction the arrangement of molecules behaves like a macroscopic magnet.*

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Ising Model

Ising model was first exploited for investigating **spontaneous magnetization** *in ferromagnetic film (i.e. magnetization in the absence of external magnetic field).*

❖ A phase transition in the context of the Ising model is a transition from an ordered state to a disordered state.

❖ A ferromagnet above the critical temperature T_c is in a disordered state. In the Ising model this corresponds to a random distribution of the spin values.

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- Below the critical temperature T_c (nearly) all spins are aligned, even in the absence of an external applied magnetic field H.

- *If we heat up a cooled ferromagnet, the magnetization vanishes at T_c and the ferromagnet switches from an ordered to a disordered state. This is a phase transition of second order.*

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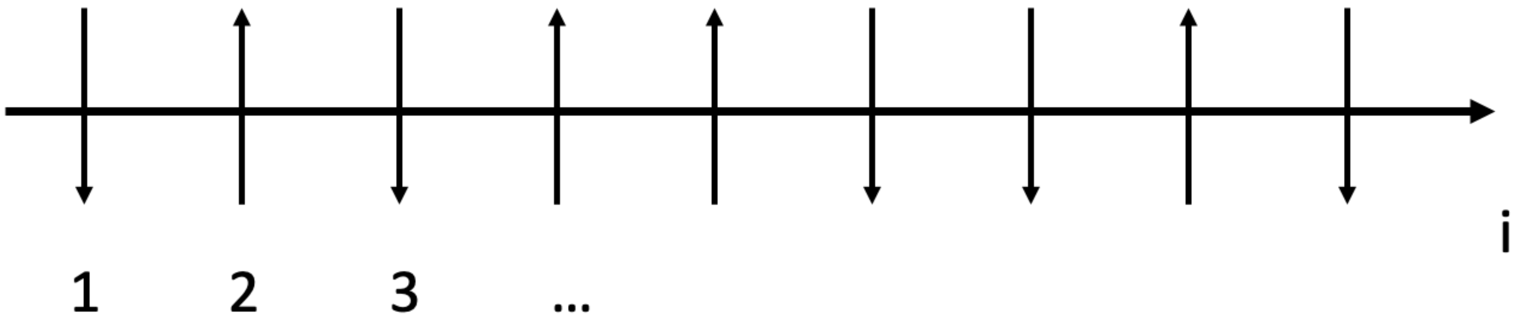
Example Consider a system of three fixed particles, each having spin $\frac{1}{2}$ so that each spin can point either up or down (i.e., along or opposite some direction chosen as the z axis). Each particle has a magnetic moment along the z axis of μ when it points up, and $-\mu$ when it points down. The system is placed in an external magnetic field H pointing along this z axis.

Example Consider a system of three fixed particles, each having spin $\frac{1}{2}$ so that each spin can point either up or down (i.e., along or opposite some direction chosen as the z axis). Each particle has a magnetic moment along the z axis of μ when it points up, and $-\mu$ when it points down. The system is placed in an external magnetic field H pointing along this z axis.

The state of the particle i can be specified by its magnetic quantum number m_i which can assume the two values $m_i = \pm\frac{1}{2}$. The state of the whole system is specified by giving the values of the three quantum numbers m_1, m_2, m_3. A particle has energy $-\mu H$ when its spin points up, and energy μH when its spin points down.

State index r	*Quantum numbers* m_1, m_2, m_3	*Total magnetic moment*	*Total energy*
1	$+ + +$	3μ	$-3\mu H$
2	$+ + -$	μ	$-\mu H$
3	$+ - +$	μ	$-\mu H$
4	$- + +$	μ	$-\mu H$
5	$+ - -$	$-\mu$	μH
6	$- + -$	$-\mu$	μH
7	$- - +$	$-\mu$	μH
8	$- - -$	-3μ	$3\mu H$

One dimensional Ising Model



❖ *Each spin can interact with its neighbors.*
❖ *No external magnetic field is considered.*

❖ The interaction strength between two spins σ_i and σ_{i+1} is characterized by the coupling strength J. The Hamiltonian $\mathcal{H}$ of such a system is then given by

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j \,,$$

with a nearest neighbouring sum.

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- Our end goal is to find various **thermodynamic properties** of the 1D Ising model.
- Remember that thermodynamics means that the system is in *thermal equilibrium*; where the probability of each configuration/microstate is $\frac{e^{-\beta E}}{Z}$ (This also called the "canonical ensemble").
- Our system is defined by a **Hamiltonian** $\mathcal{H}$ a function that tells us its energy. From the Hamiltonian, we figure out the **energy eigenstates**, a.k.a. **configurations**, (or sometimes even "states") of the system. These are the basic states, and we will be *summing over* these states.
- We take limit $N \to \infty$

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One dimensional Ising Model

In the canonical ensemble, the probability of finding a particular spin configuration $\{s_i\}$ is,

$$p(\{s_i\}) = \frac{1}{Z}\exp(-\beta H(\{s_i\})), \quad \beta \equiv \frac{1}{k_B T} \tag{2}$$

where $Z = \sum_{\{s_i\}} \exp(-\beta H(\{s_i\}))$ is the partition function. Due to the Boltzmann factor, $e^{-\beta H}$, spin configurations with lower energies will be favored.

states, and we will be *summing over* these states.

- We take limit $N \to \infty$
- And our main target is to calculate partition functions $Z = \sum e^{-\beta E(s)}$, β is the related to inverse temp, and $E(s)$ is the energy of the system when it is in state s.
- Once we have found Z, we can calculate the free energy $F = -T \log Z$

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One dimensional Ising Model

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- For a system with N_{tot} lattice sites, and two possible $\sigma_i -$ values, a total number of $2^{N_{\text{tot}}}$ possible configurations of the arrangements of particles exist. Summing over all configurations, *our main target* is obtain the partition sum Z as

$$Z = \sum_{\{i\}} e^{-\beta \mathcal{H}} = \sum_{\sigma_1 = \pm 1} \sum_{\sigma_2 = \pm 1} \dots \dots . \sum_{\sigma_N = \pm 1} e^{\beta J (\sigma_1 \sigma_2 + \sigma_2 \sigma_3 + \cdots .)}$$

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We introduce a new variable $\mu_i \coloneqq \sigma_i . \sigma_{i+1}$

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

Thus, for $N \gg 1$, $\quad Z = 2. \sum_{\{\mu\}} e^{\beta J \sum_{i=1}^{N-1} \mu_i}$

The factor of 2 in the partition function arises from the two possible configurations for the first spin in the chain.

One dimensional Ising Model

$$\mathcal{H} = -J\sum_{\langle ij\rangle} \sigma_i\sigma_j \qquad Z = \sum_{\{i\}} e^{-\beta\mathcal{H}} = \sum_{\sigma_1=\pm1}\sum_{\sigma_2=\pm1} \ldots\ldots \sum_{\sigma_N=\pm1} e^{\beta J(\sigma_1\sigma_2+\sigma_2\sigma_3+\cdots.)}$$

$$Z = 2.\sum_{\{\mu\}} e^{\beta J\sum_{i=1}^{N-1}\mu_i} = 2\sum_{\mu_1=\pm1}\sum_{\mu_2=\pm1}\ldots..\sum_{\mu_{N-1}=\pm1} e^{\beta J(\mu_1+\mu_2+\mu_3+\cdots.+\mu_{N-1})}$$

Open Boundary condition

$$= 2\sum_{\mu_1=\pm1}\sum_{\mu_2=\pm1}\ldots..\sum_{\mu_{N-2}=\pm1} e^{\beta J(\mu_1+\mu_2+\mu_3+\cdots.+\mu_{N-2})} \underbrace{\sum_{\mu_{N-1}}}_{e^{\beta J}+e^{-\beta J}}$$

$$= 2[2\cosh(\beta J)]^{N-1} \approx [2\cosh(\beta J)]^N$$

$$\begin{aligned}
\cosh x &= \frac{e^x + e^{-x}}{2} \\
\sinh x &= \frac{e^x - e^{-x}}{2} \\
\frac{d}{dx}\cosh x &= \sinh x \\
\frac{d}{dx}\sinh x &= \cosh x \\
\tanh x &= \frac{\sinh x}{\cosh x} \\
\frac{d}{dx}\tanh x &= 1 - (\tanh x)^2
\end{aligned}$$

This is the partition function of the one-dimensional Ising model without an external field.

$$F = -kT \ln Z_N$$

$$S = Nk\big[\ln(e^{2\beta J} + 1) - \frac{2\beta J}{1 + e^{-2\beta J}}\big].$$

$$E = -NJ \tanh \beta J.$$

$$C = Nk(\beta J)^2 (\operatorname{sech} \beta J)^2.$$

One dimensional Ising Model

Reason for learning Ising model in 1-D.
Strategy: for a given Hamiltonian construct the partition function.
Weakness: Ising;s 1-d solution was unable to find the phase transition.

Q: **What will happen if we add external magnetic field? How can we calculate that?**

One dimensional Ising Model:Transfer matrix

The trace is the sum of the diagonal elements of a matrix

$$\mathrm{Tr}(B) = B_{11} + B_{22} + \cdots + B_{nn}$$

$$\text{where} \quad B = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & B_{22} & \cdots & B_{2n} \\ \vdots & \vdots & \ddots & \\ B_{n1} & B_{n2} & \cdots & B_{nn} \end{pmatrix}$$

'or an Ising model with one spin, $H = -h\, s_1$, the partition function is

$$Z = \mathrm{Tr}\begin{pmatrix} e^{\beta h} & 0 \\ 0 & e^{-\beta h} \end{pmatrix} = e^{\beta h} + e^{-\beta h}$$

Wei cai SU

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Now consider two spins, under periodic B.C.,

$$H(s_1, s_2) = -J s_1 s_2 - J s_2 s_1 - h s_1 - h s_2$$

Define matrix

$$P \equiv \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$

then,

$$Z = \sum_{\{s_1 s_2\}} e^{-\beta H} = \mathrm{Tr}(P \cdot P)$$

One dimensional Ising Model:Transfer matrix

Let the 1st row (column) correspond to $s = +1$, and let the 2nd row(column) correspond to $s = -1$, e.g.,

$$\begin{aligned} P_{+1,+1} &= \mathrm{e}^{\beta(J+h)}, & P_{+1,-1} &= \mathrm{e}^{-\beta J}, \\ P_{-1,+1} &= \mathrm{e}^{-\beta J}, & P_{-1,-1} &= \mathrm{e}^{\beta(J-h)}, \end{aligned}$$

$$P_{s_1 s_2} = \mathrm{e}^{\beta\left(J s_1 s_2 + \frac{h}{2} s_1 + \frac{h}{2} s_2\right)}$$

$$\mathrm{Tr}(P \cdot P) = \sum_{s_1} (P \cdot P)_{s_1, s_1} = \sum_{s_1, s_2} P_{s_1, s_2} P_{s_2, s_1}$$

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In general, for N spins forming a linear chain under PBC, the partition function is

$$\begin{aligned} Z &= \sum_{\{s_i\}} \mathrm{e}^{-\beta H(\{s_i\})} \\ &= \sum_{\{s_i\}} \mathrm{e}^{\beta\left(J s_1 s_2 + \frac{h}{2} s_1 + \frac{h}{2} s_2\right)} \cdot \mathrm{e}^{\beta\left(J s_2 s_3 + \frac{h}{2} s_2 + \frac{h}{2} s_3\right)} \cdots \\ &\quad \mathrm{e}^{\beta\left(J s_{N-1} s_N + \frac{h}{2} s_{N-1} + \frac{h}{2} s_N\right)} \cdot \mathrm{e}^{\beta\left(J s_N s_1 + \frac{h}{2} s_N + \frac{h}{2} s_1\right)} \\ &= \mathrm{Tr}\left(P^N\right) \end{aligned}$$

One dimensional Ising Model:Transfer matrix

1. Every symmetric (real) matrix can be diagonalized,

$$P = U \cdot D \cdot U^T \tag{27}$$

where U is a unitary matrix ($U \cdot U^T = I$), and D is a diagonal matrix. For 2×2 matrices, define $\lambda_+ \equiv D_{11}$, $\lambda_- \equiv D_{22}$ ($D_{12} = D_{21} = 0$). $\lambda_\pm$ are the eigenvalues of matrix P.

2. Trace is unchanged after diagonalization

$$\mathrm{Tr(P)} = \mathrm{Tr(D)} = \lambda_+ + \lambda_- \tag{28}$$

Hence the trace equals the sum of the eigenvalues.

3. The same matrix U that diagonalizes P also diagonalizes P^N, because

$$P^N = (U \cdot D \cdot U^T) \cdot (U \cdot D \cdot U^T) \cdots (U \cdot D \cdot U^T) = U \cdot D^N \cdot U^T \tag{29}$$

4. Notice that

$$D^N = \begin{pmatrix} \lambda_+^N & 0 \\ 0 & \lambda_-^N \end{pmatrix} \tag{30}$$

We have

$$\mathrm{Tr}(P^N) = \mathrm{Tr}(D^N) = \lambda_+^N + \lambda_-^N \tag{31}$$

One dimensional Ising Model:Transfer matrix

$$\lambda_\pm = e^{\beta J}\left[\cosh\beta h \pm \sqrt{\sinh^2\beta h + e^{-4\beta J}}\right]$$

$$U = \begin{bmatrix} -e^{\beta J}\left(e^{\beta(J-h)-\lambda_+}\right) & 1 \\ 1 & -e^{\beta J}\left(e^{\beta(J+h)-\lambda_-}\right) \end{bmatrix}$$

$$\mathrm{Tr}(P) = \lambda_+ + \lambda_- = 2\,e^{\beta J}\cosh\beta h$$

$$\begin{aligned} Z &= \mathrm{Tr}(P^N) = \lambda_+^N + \lambda_-^N \\ &= e^{N\beta J}\left\{\left[\cosh\beta h + \sqrt{\sinh^2\beta h + e^{-4\beta J}}\right]^N \right. \\ &\quad \left. + \left[\cosh\beta h - \sqrt{\sinh^2\beta h + e^{-4\beta J}}\right]^N\right\} \end{aligned}$$

h and H (used in class) are same

One dimensional Ising Model:Transfer matrix

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$$\frac{1}{N}\ln Z_N(T,H) = \ln\lambda_+ + \ln\left[1+\left(\frac{\lambda_-}{\lambda_+}\right)^N\right] \underset{N\to\infty}{\longrightarrow} \ln\lambda_+,$$

$$\begin{aligned} Z &= \mathrm{Tr}(P^N) = \lambda_+^N + \lambda_-^N \\ &= e^{N\beta J}\left\{\left[\cosh\beta h + \sqrt{\sinh^2\beta h + e^{-4\beta J}}\right]^N \right. \\ &\quad \left. + \left[\cosh\beta h - \sqrt{\sinh^2\beta h + e^{-4\beta J}}\right]^N\right\} \end{aligned}$$

Harvey Gould and Jan Tobochnik

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and the free energy per spin is given by

$$\frac{1}{N}F(T,H) = -kT\ln\left[e^{\beta J}\cosh\beta H + \left(e^{2\beta J}\sinh^2\beta H + e^{-2\beta J}\right)^{1/2}\right].$$

$$M = \frac{\partial F}{\partial H} = N\frac{\sinh\beta H}{(\sinh^2\beta H + e^{-4\beta J})^{1/2}}.$$

Harvey Gould and Jan Tobochnik

One dimensional Isina Model:Transfer matrix

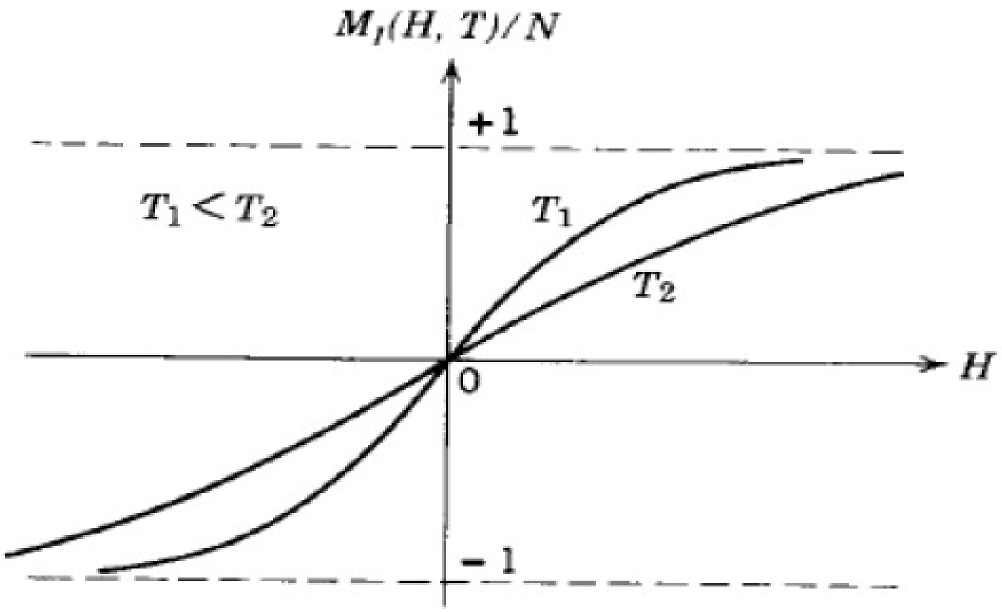


Fig. 14.16 Magnetization of the one-dimensional Ising model. There is no spontaneous magnetization.

$$M = \frac{\partial F}{\partial H} = N \frac{\sinh \beta H}{(\sinh^2 \beta H + e^{-4\beta J})^{1/2}}.$$

A system is paramagnetic if $M \neq 0$ only for $H \neq 0$, and is ferromagnetic if $M \neq 0$ for $H = 0$. For the one-dimensional Ising model, we see from (5.72) that $M = 0$ for $H = 0$, and there is no spontaneous magnetization at nonzero temperature. (Recall that $\sinh x \approx x$ for small x.) That is, the one-dimensional Ising model undergoes a phase transition from the paramagnetic to the ferromagnetic state only at $T = 0$. In the limit of low temperature ($\beta J \gg 1$ and $\beta H \gg 1$), $\sinh \beta H \approx \frac{1}{2} e^{\beta H} \gg e^{-2\beta J}$ and $m = M/N \approx 1$ for $H \neq 0$. Hence, at low temperatures only a small field is needed to produce saturation, corresponding to $m = 1$.

Phase Transition in 2-d Ising model of a Ferromagnet

The two-dimensional Ising model was solved exactly in zero magnetic eld for a rectangular lattice by Lars Onsager in 1944. Onsager's calculation was the first exact solution that exhibited a phase transition in a model with short-range interactions. Before his calculation, some people believed that statistical mechanics was not capable of yielding a phase transition.

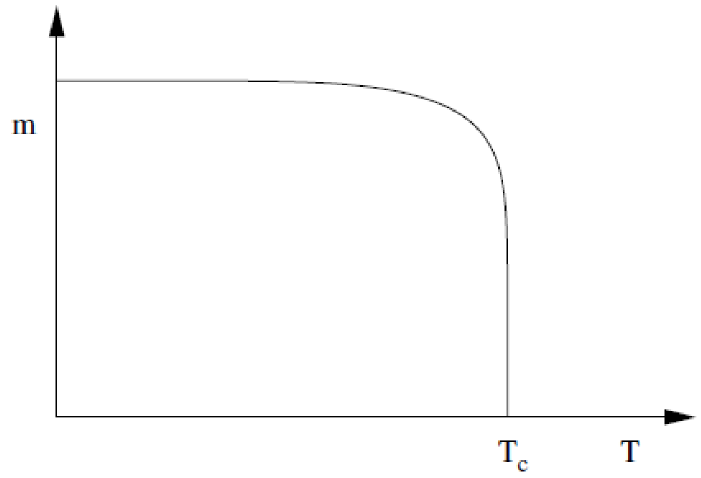


Figure 5.8: The temperature-dependence of the spontaneous magnetization of the two-dimensional Ising model.

The most important results of the exact solution of the two-dimensional Ising model are that the energy (and the free energy and the entropy) are continuous functions for all T, m vanishes continuously at $T = T_c$, the heat capacity diverges logarithmically at $T = T_c$, and the zero-field susceptibility diverges as a power law. When we discuss phase transitions in more detail in Chapter 9, we will understand that the paramagnetic ↔ ferromagnetic transition in the two-dimensional Ising model is *continuous*. That is, the order parameter m vanishes continuously rather than discontinuously. Because the transition occurs only at $T = T_c$ and $H = 0$, the transition occurs at a *critical point*.

Phase Transition in 2-d Ising model of a Ferromagnet

In most ordinary materials the associated magnetic dipoles of the atoms have a random orientation. In effect this non-specific distribution results in no overall macroscopic magnetic moment. However in certain cases, such as iron, a magnetic moment is produced as a result of a preferred alignment of the atomic spins.

This phenomenon is based on two fundamental principles, namely energy minimization and entropy maximization.

These are competing principles and are important in moderating the overall effect. Temperature is the mediator between these opposing elements and ultimately determines which will be more dominant

Jacques Kotze
arXiv:0803.0217v1

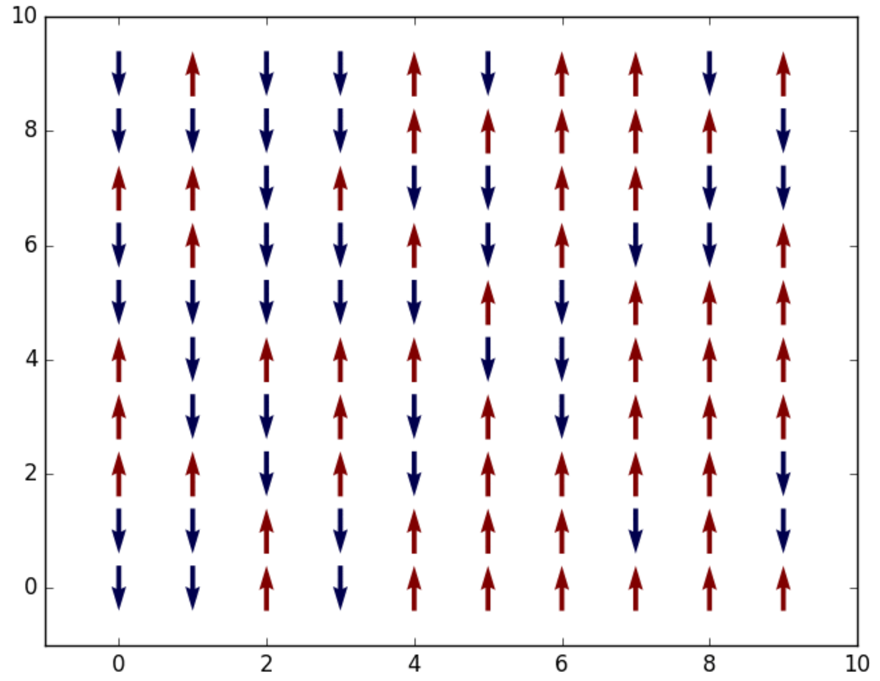
Phase Transition in 2-d Ising model of a Ferromagnet

Aim: Critical Temperature for phase transition

Let $s_{i,j}$ denote a spin state at lattice coordinates i and j having either spin up or spin down, $s_{i,j} = \pm 1$. The Hamiltonian or total energy of the system in a particular state $\{s_{i,j}\}$ is

$$H(\{s_{i,j}\}) = -J \sum_{i,j} s_{i,j}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1})$$

Here, in 2D Ising model, the spins are lying in xy-plane and oriented only along either $+y$ or $-y$ directions.



https://ruihaoqiu.github.io/MC-Magnetic-Phase-Transition/

Phase Transition in 2-d Ising model of a Ferromagnet

Aim: *Critical Temperature for phase transition*

Let $s_{i,j}$ denote a spin state at lattice coordinates i and j having either spin up or spin down, $s_{i,j} = \pm 1$. The Hamiltonian or total energy of the system in a particular state $\{s_{i,j}\}$ is

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Periodic boundary condition (2 d lattice folded in torus) and only a nearest neighbour coupling is allowed.

The system of spins is considered to be in contact with a heat bath at temperature T. The set of spins can exchange energy with the heat bath, in such a way that the system can come into thermal equilibrium with the heat bath. In thermal equilibrium, the probability of finding the system in a particular microstate α is :

$$P_\alpha \propto e^{-\frac{E_\alpha}{kT}}$$

Phase Transition in 2-d Ising model of a Ferromagnet

Aim: Critical Temperature for phase transition

Let $s_{i,j}$ denote a spin state at lattice coordinates i and j having either spin up or spin down, $s_{i,j} = \pm 1$. The Hamiltonian or total energy of the system in a particular state $\{s_{i,j}\}$ is

$$H(\{s_{i,j}\}) = -J \sum_{i,j} s_{i,j}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1})$$

If the energy of each possible state of the system is specified, then the Boltzmann distribution function, gives the probability for the system to be in each possible state (at a given temperature) and therefore macroscopic quantities of interest can be calculated by doing probability summing. This can be illustrated by using magnetization and energy as examples. For any fixed state α, the magnetization is proportional to the 'excess' number of spins pointing up or down while the energy is given by the Hamiltonian H

$$M(\alpha) = N_{up}(\alpha) - N_{down}(\alpha)$$

Jacques Kotze

Phase Transition in 2-d Ising model of a Ferromagnet

Aim: Critical Temperature for phase transition

Let $s_{i,j}$ denote a spin state at lattice coordinates i and j having either spin up or spin down, $s_{i,j} = \pm 1$. The Hamiltonian or total energy of the system in a particular state $\{s_{i,j}\}$ is

$$H(\{s_{i,j}\}) = -J \sum_{i,j} s_{i,j}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1})$$

$$M(\alpha) = N_{up}(\alpha) - N_{down}(\alpha)$$

Thus, expected value $\langle M \rangle = \sum_{\alpha} M(\alpha) P_{\alpha}$. similarly, $\langle E \rangle = \sum_{\alpha} E(\alpha) P_{\alpha}$

These calculation pose a drastic problem from a practical perspective. Considering we have two spin orientations (up & down) and there are N spins which implies that there are 2^N different states. As N becomes large it is evident that computations become a daunting task if calculated in this manner.

Jacques Kotze

Quantum Mechanics of Magnetism Summarized

- Due to the motion of their electrons, some atoms can have intrinsic magnetic moments. Other atoms develop magnetic moments when placed in an external magnetic field.

- If atoms have zero magnetic moment, then an applied field H induces magnetic moments that are aligned anti-parallel to the applied field. The magnetization M is such that $M/H < 0$. This is **diamagnetism.**

- If atoms have non-zero magnetic moments that point in random directions then the sum over many atoms gives zero magnetization. An applied field tends to align the magnetic moments so that $M/H > 0$. This is **paramagnetism.**

- If atoms have non-zero magnetic moments that point in the same direction then the sum over many atoms gives finite magnetization. This is **ferromagnetism.**

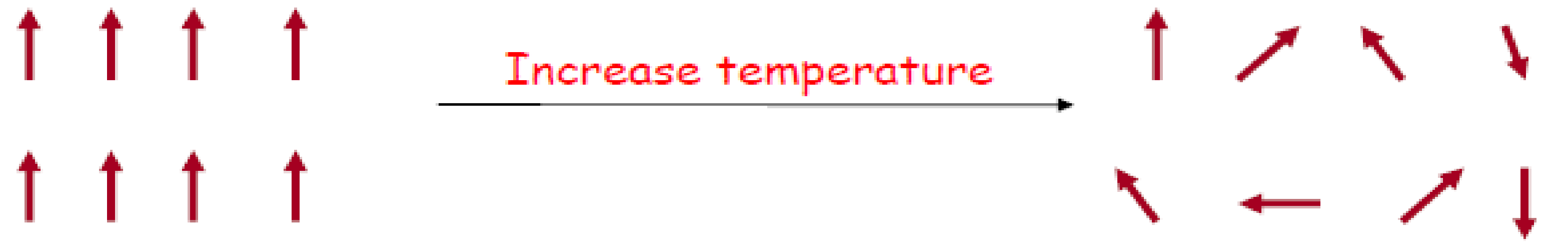
Bohr-van Leeuwen Theorem: At any finite temperature, and in all finite applied electric or magnetic fields, classical magnetization of a collection of electrons in thermal equilibrium vanishes identically.

Branislav K. Nikolić

Magnetization as Order Parameter of Alligned Quantum Spins

Ferromagnet:
Spins aligned

Paramagnet:
Spins disordered



- Spontaneous magnetization (in the absence of magnetic field): 'Order Parameter'.
- Ordered phase spontaneously breaks spin rotation symmetry.
- **First Order Phase Transitions:** discontinuity in the order parameter or energy (i.e., first derivative of the free energy).
- **Second order Phase Transitions:** divergence in the susceptibility or specific heat (i.e., second derivative of the free energy).

Branislav K. Nikolić

How to Evade **Analytic** Computation of Ising Model Partition Function?

- Suppose we want to calculate the exact partition function **Z** numerically. Need to do this for all **T**, but let us start with just one temperature.

- Real system has $O(10^{23})$ spins, try first 32x32 lattice with $O(10^3)$ spins:

  → Number of configurations in the sum = $2^{32\times 32} \sim 10^{300}$

  → Gigantic parallel supercomputer with 10 million processors: Each processor could generate a configuration ***C***, calculate its energy **E(*C*)** and the Boltzmann factor **exp[-E(*C*)/kT]**, and add it to the sum over configurations in one **nanosecond**.

  → The calculation runs during the whole age of the Universe:

$$10^7\,\mathrm{CPU} \times 10^9\,\frac{\text{configurations}}{\mathrm{CPU}\times\mathrm{s}} \times 10^{14}\,\frac{\mathrm{sec}}{\mathrm{year}}\,10^{10}\ \text{years}$$

$\sim 10^{40}$ (just for a single temperature!)

Theory Behind Metropolis Monte Carlo: Markov Chains and Detailed Balance

- Let us set up a so-called Markov chain of configurations by the introduction of a fictitious dynamics → the "time" t is computer time (marking the number of iterations of the procedure), NOT real time since our statistical system is considered to be in equilibrium, and therefore time invariant.

  →Probability to be in configuration A at time T is: $P(A,t)$

  →The transition probability (per unit time) to go from A to B is: $W(A \to B)$

$$P(A,t+1) = P(A,t) + \sum_B [\underbrace{W(B \to A)P(B,t) - W(A \to B)P(A,t)}_{=0}]$$

- Equilibrium $\lim_{t\to\infty} P(A,t) = p(A)$ can be ensured by satisfying detailed balance
- For the special case of the Boltzmann probability distribution $p(A) = e^{-E(A)/kT}/Z$

$$\frac{W(A \to B)}{W(B \to A)} = \frac{p(B)}{p(A)} = \frac{e^{-E(B)/kT}}{e^{-E(A)/kT}} = e^{-\Delta E/kT}$$

$$\Delta E = E(B) - E(A)$$

Over Los Alamos dinner party in 1953

Metropolis, Teller, and Rosenbluth chose:

$$W(A \to B) = \begin{cases} e^{-\Delta E/kT} & , \Delta E > 0 \\ 1 & , \Delta E \le 0 \end{cases}$$

Phase transition: computation(2 d lattice)

0. Start with some spin configuration $\{s_i\}$.
1. Randomly choose a spin s_i
2. Attempt to flip it, i.e. $s_i := -s_i$ (trial).
3. Compute the energy change ΔE due to this flip.
4. If $\Delta E < 0$, accept the trial.
5. If $\Delta E > 0$, accept the trial with probability $p^{\text{acc}} = e^{-\beta \Delta E}$.
6. If trial is rejected, put the spin back, i.e. $s_i := -s_i$.
7. Go to 1, unless maximum number of iterations is reached.

$$H(\{s_{i,j}\}) = -J \sum_{i,j} s_{i,j}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1})$$

Wei Cai Stanford University

Phase transition: computation(2 d lattice)

```
adj = neighbors(1:N^2,N);
% grid = sign(randn(N,N));
p=.5; % average proportion of initial +1 spins
grid = sign(p-rand(N));
```

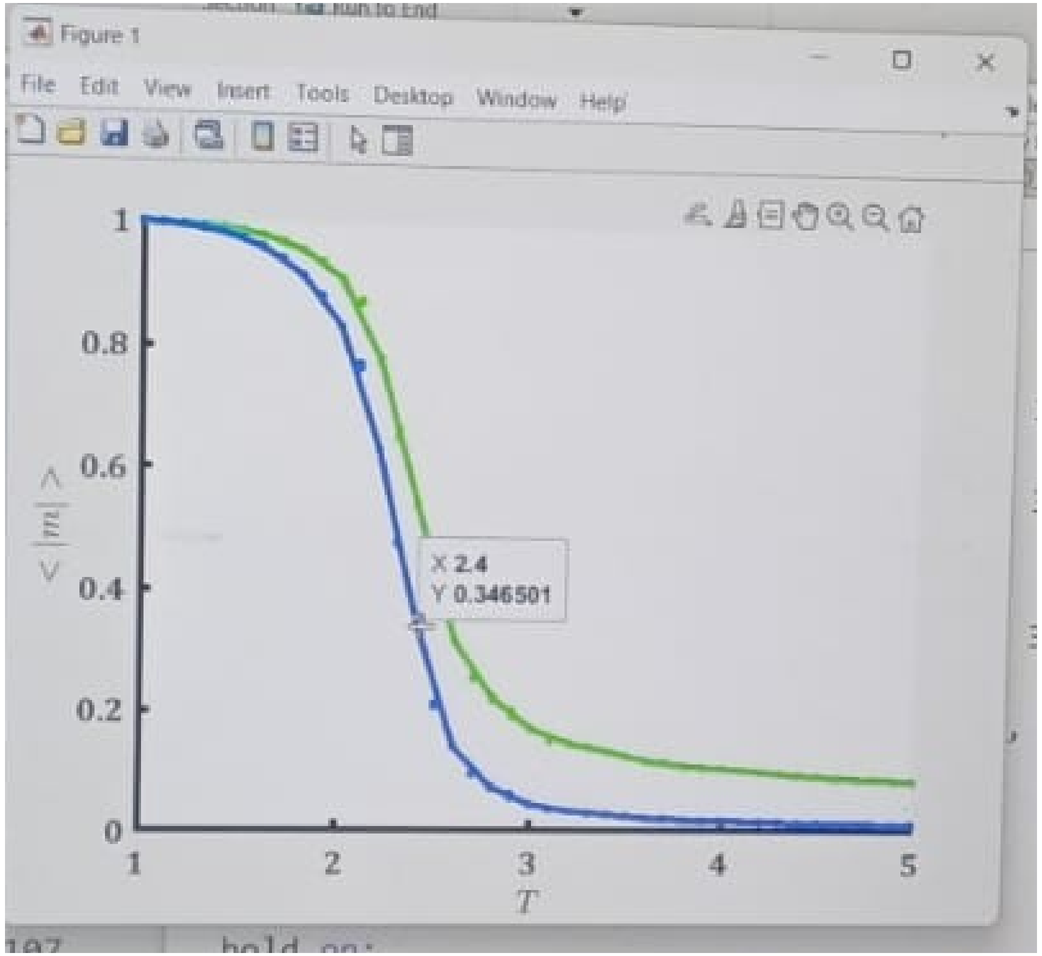
Phase transition: computation(2 d lattice)

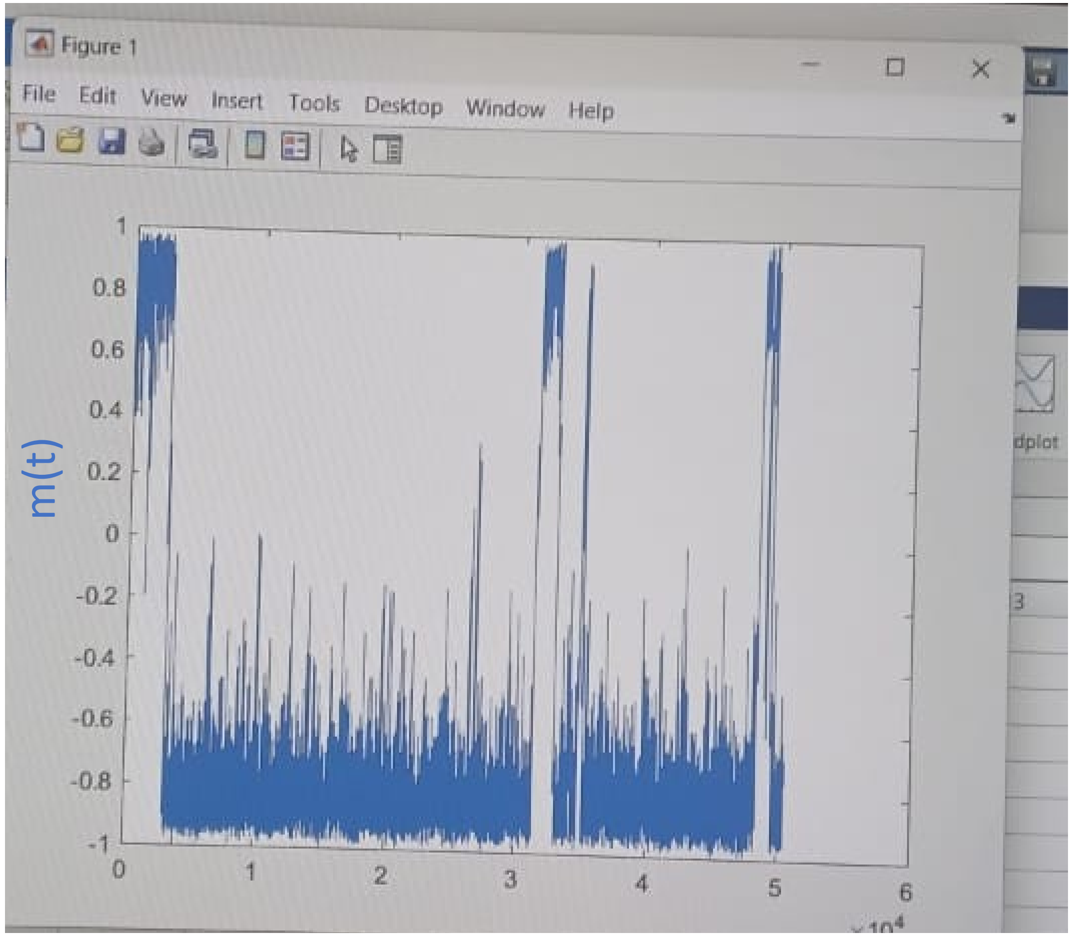
```
adj = neighbors(1:N^2,N);
% grid = sign(randn(N,N));
p=.5; % average proportion of initial +1 spins
grid = sign(p-rand(N));
```

```
for T = 1.57:0.1:5;
kT = 1*T;
s=randi(N^2);
% location of ith spin
% s = spin(i);
% Calculate the change in energy of flipping s
dE = 2*J*grid(s)*sum(grid(adj(s,:)));
% Calculate the transition probability
p = exp(-dE/kT);
% Decide if a transition will occur
if dE<=0 || rand <= p,
grid(s) = -1*grid(s);
end
```

$$H(\{s_{i,j}\}) = -J \sum_{i,j} s_{i,j}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1})$$

Phase transition: computation(2 d lattice)





Mc steps

Phase transition: computation(2 d lattice)

Q: Why does the Metropolis algorithm generate the canonical distribution?

To simplify the notation, let A, B represent arbitrary spin configurations $\{s_i\}$. Our goal is to prove that when the MC simulation has reached equilibrium, the probability of sampling state A is

$$p_A = \frac{1}{Z} e^{-\beta H(A)} \tag{52}$$

where

$$Z = \sum_A e^{-\beta H(A)} \tag{53}$$

— the sum is over all possible (2^N) spin configurations.

Monte Carlo simulation follows a Markov Chain, which is completely specified by a transition probability matrix π_{AB} — the probability of jumping to state B in the next step if the current state is A.

For an Ising model with N spins, there are 2^N spin configurations (states). So π_{AB} is a $2^N \times 2^N$ matrix. However, most entries in π_{AB} are zeros.

$\pi_{AB} \neq 0$ only if there is no more than one spin that is different (flipped) between A and B. For example,

$$\begin{aligned}
\text{if } A &= \{+1,+1,+1,+1,+1,+1\} \quad \text{then} \\
\text{for } B &= \{+1,-1,+1,+1,+1,+1\}, \quad \pi_{AB} > 0 \\
\text{but for } B &= \{-1,-1,+1,+1,+1,+1\}, \quad \pi_{AB} = 0
\end{aligned}$$

Wei Cai Stanford University

Phase transition: computation(2 d lattice)

To prove the Metropolis algorithm generates the canonical ensemble:

(1) transition matrix can be written as

$$\pi_{AB} = \alpha_{AB} \cdot p^{\mathrm{acc}}_{AB}, \quad \text{for } B \neq A \tag{54}$$

$$\pi_{AA} = 1 - \sum_{B \neq A} \pi_{AB} \tag{55}$$

where α_{AB} is the trial probability that satisfies

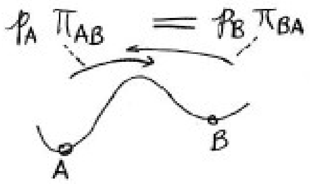
$$\alpha_{AB} = \alpha_{BA} \tag{56}$$

and p^{acc}_{AB} is the acceptance probability.

without loss of generality, let's assume $E_B > E_A$, then

$$\begin{cases} p^{\mathrm{acc}}_{AB} = \exp\left(-\frac{E_B - E_A}{k_B T}\right) \\ p^{\mathrm{acc}}_{BA} = 1 \end{cases} \implies \frac{\pi_{AB}}{\pi_{BA}} = \frac{\alpha_{AB}}{\alpha_{BA}} \frac{p^{\mathrm{acc}}_{AB}}{p^{\mathrm{acc}}_{BA}} = \exp\left(-\frac{E_B - E_A}{k_B T}\right) \tag{57}$$

(2) If the equilibrium distribution is reached, with p_A being the probability of sampling state A, then we expect the following balance of fluxes.



$$\begin{aligned} \therefore \quad p_A \pi_{AB} &= p_B \pi_{BA} \\ \frac{p_A}{p_B} &= \frac{\pi_{BA}}{\pi_{AB}} = \mathrm{e}^{\frac{E_B - E_A}{k_B T}} \\ \implies \quad p_A &= \mathrm{const} \cdot \mathrm{e}^{-\frac{E_A}{k_B T}} \quad \text{for all } A \end{aligned} \tag{58}$$

The normalization of p_A, $\sum_A p_A = 1$, requires that

$$p_A = \frac{1}{Z} \mathrm{e}^{-\frac{E_A}{k_B T}}, \qquad Z = \sum_A \mathrm{e}^{-\frac{E_A}{k_B T}} \tag{59}$$

(end of proof)

Wei Cai Stanford University